

Systems Design: Process Costing

Chapter Four



Similarities Between Job-Order and Process Costing

- Both systems assign material, labor and overhead costs to products and they provide a mechanism for computing unit product cost.
- Both systems use the same manufacturing accounts, including Manufacturing Overhead, Raw Materials, Work in Process, and Finished Goods.
- The flow of costs through the manufacturing accounts is basically the same in both systems.

Differences Between Job-Order and Process Costing

- Process costing is used when a single product is produced on a continuing basis or for a long period of time. Job-order costing is used when many different jobs are worked in each period.
- Process costing systems accumulate costs by department. Job-order costing systems accumulate costs by individual jobs.
- Process costing systems compute unit costs by department. Job-order costing systems compute unit costs by job on the job cost sheet.

Quick Check ✓

Process costing is used for products that are:

- a. Different and produced continuously.
- b. Similar and produced continuously.
- c. Individual units produced to customer specifications.
- d. Purchased from vendors.

Processing Departments

Any unit in an organization where materials, labor or overhead are added to the product.

The activities performed in a processing department are *performed uniformly* on all units of production. Furthermore, the output of a processing department must be *homogeneous*.



Production Process of Coca-cola

cleaning

mixing
& blending

bottling

capping

labeling

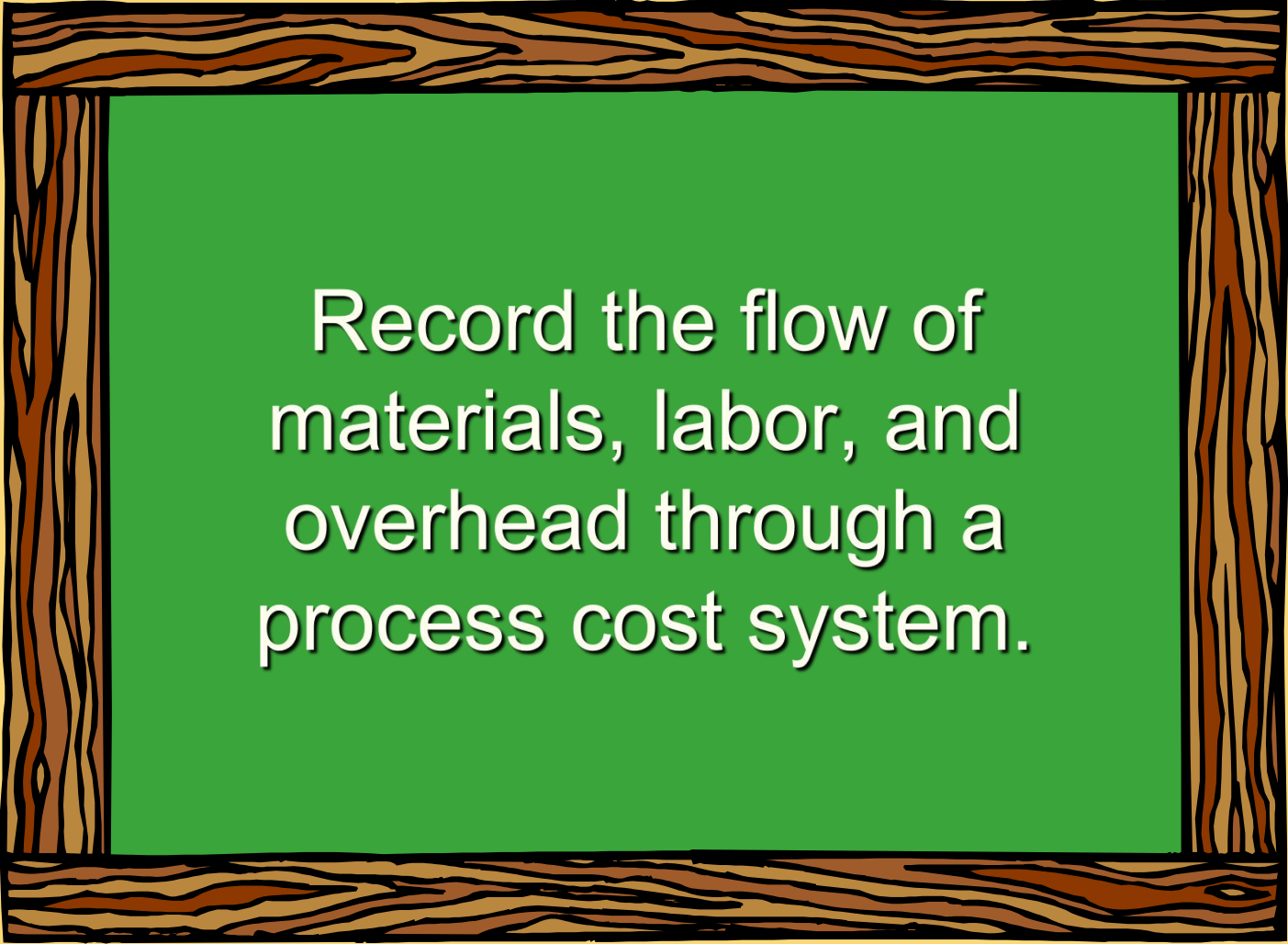
coding

quality
checking

packaging

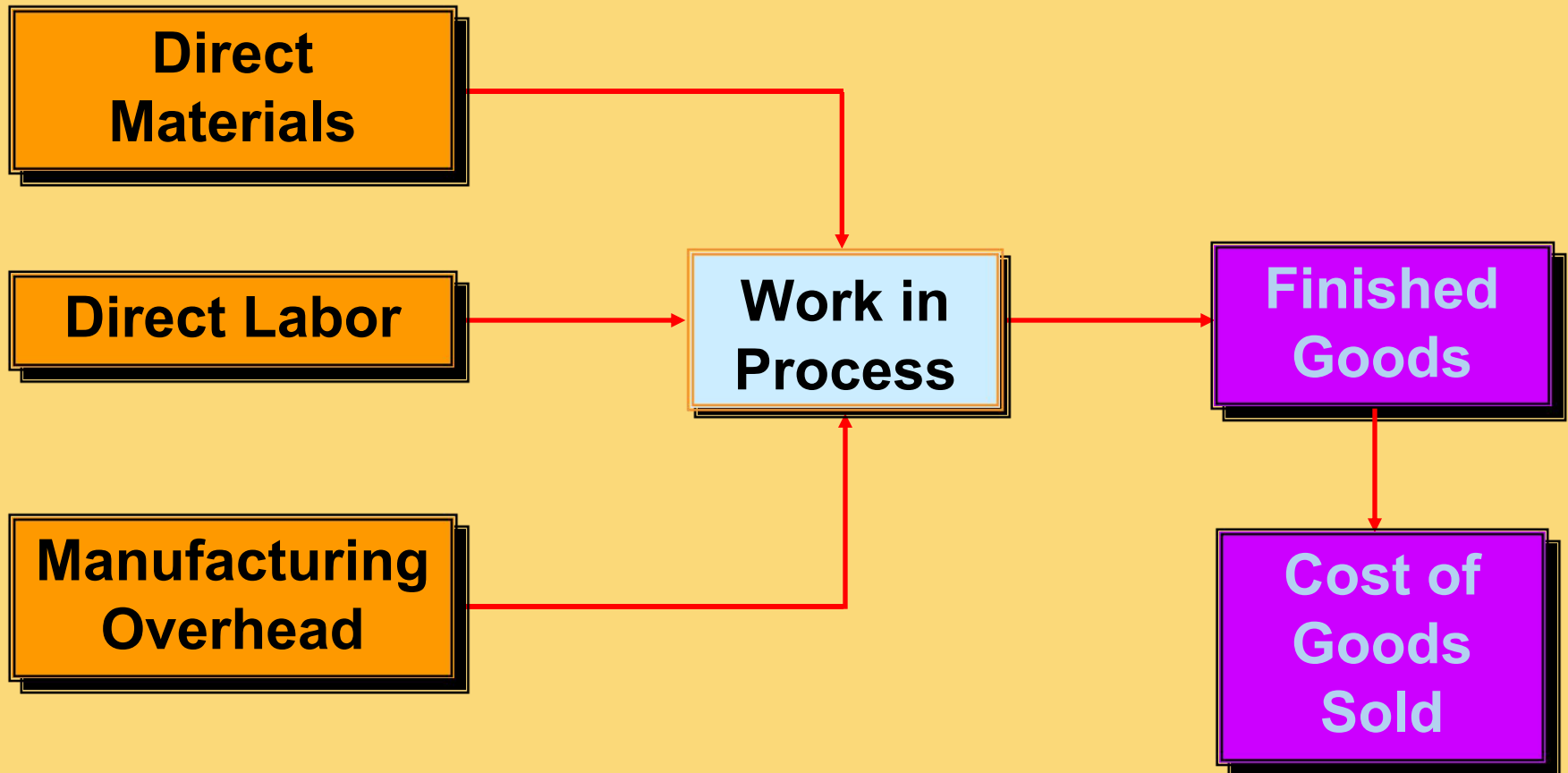


Learning Objective 1

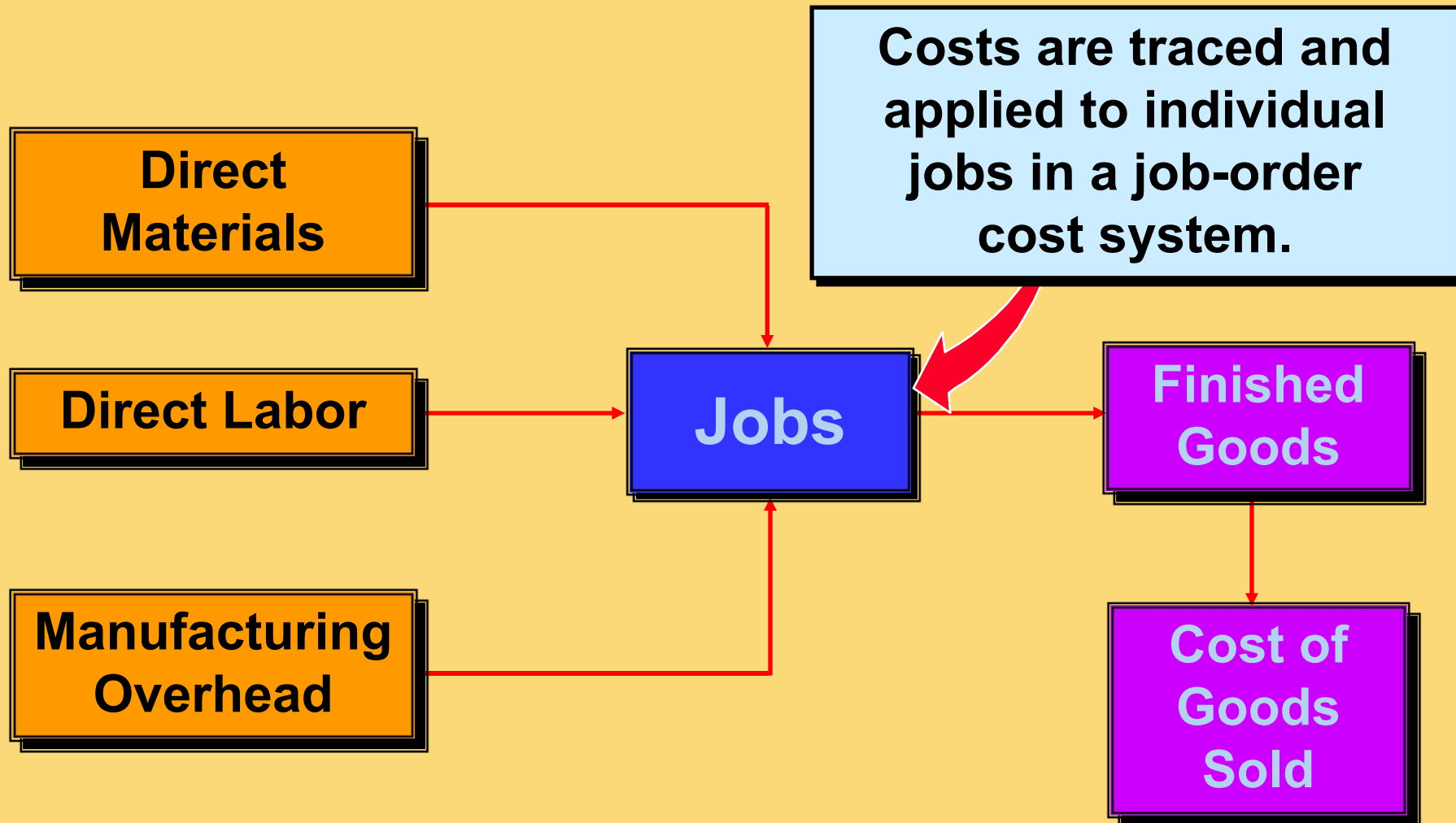


Record the flow of
materials, labor, and
overhead through a
process cost system.

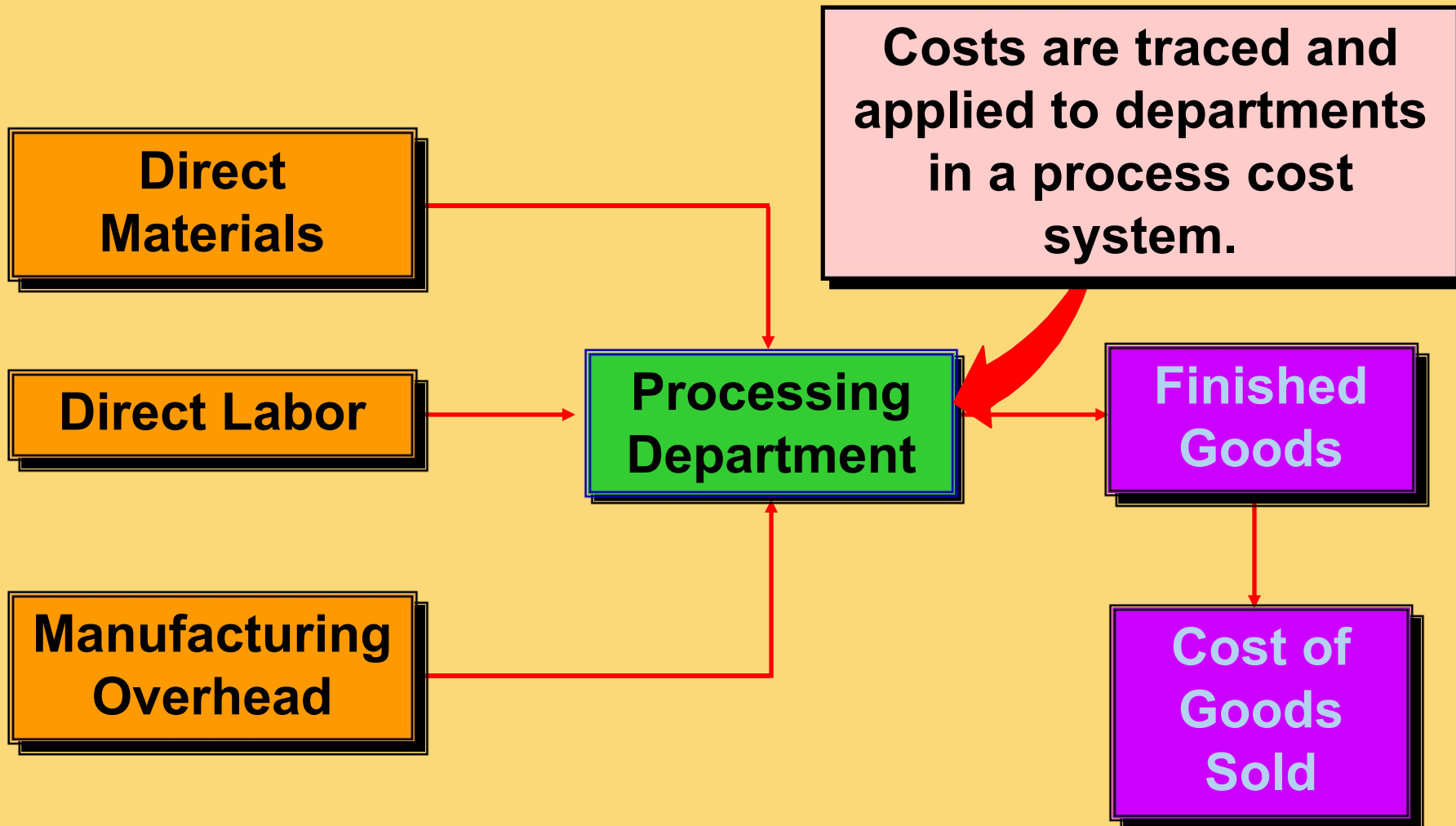
Comparing Job-Order and Process Costing



Comparing Job-Order and Process Costing

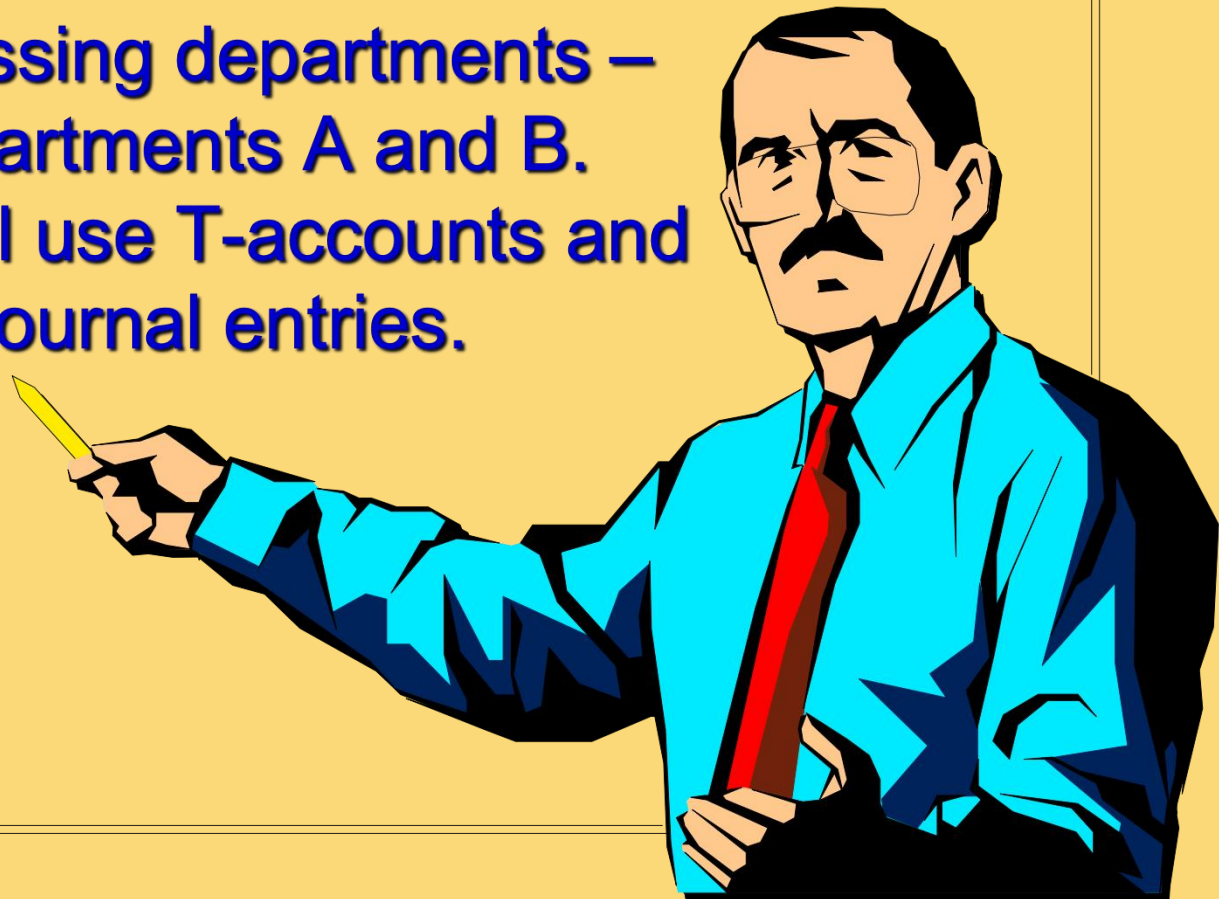


Comparing Job-Order and Process Costing

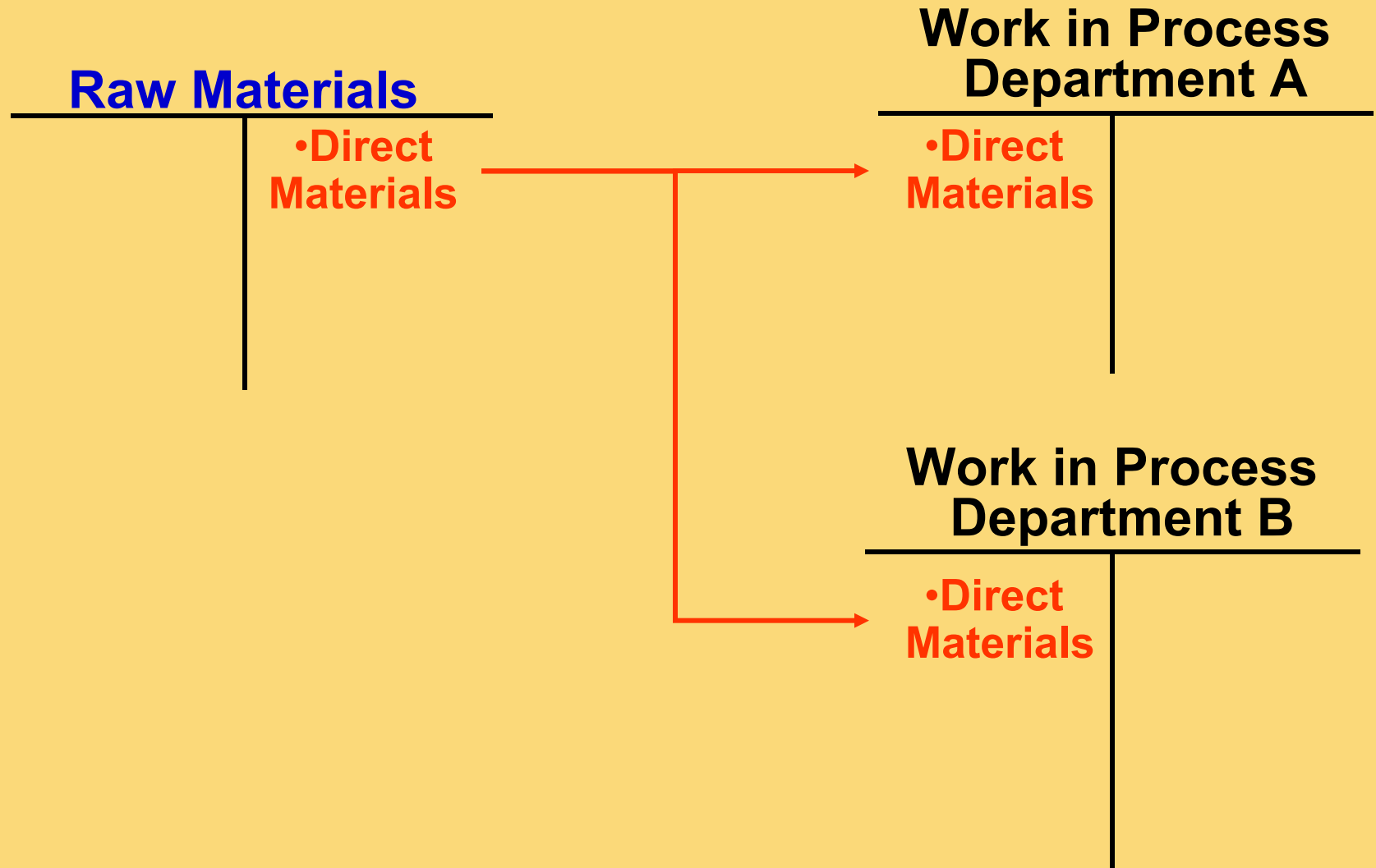


T-Account and Journal Entry Views of Cost Flows

For purposes of this example,
assume there are two
processing departments –
Departments A and B.
We will use T-accounts and
journal entries.



Process Cost Flows (in T-account form)



Process Cost Flows

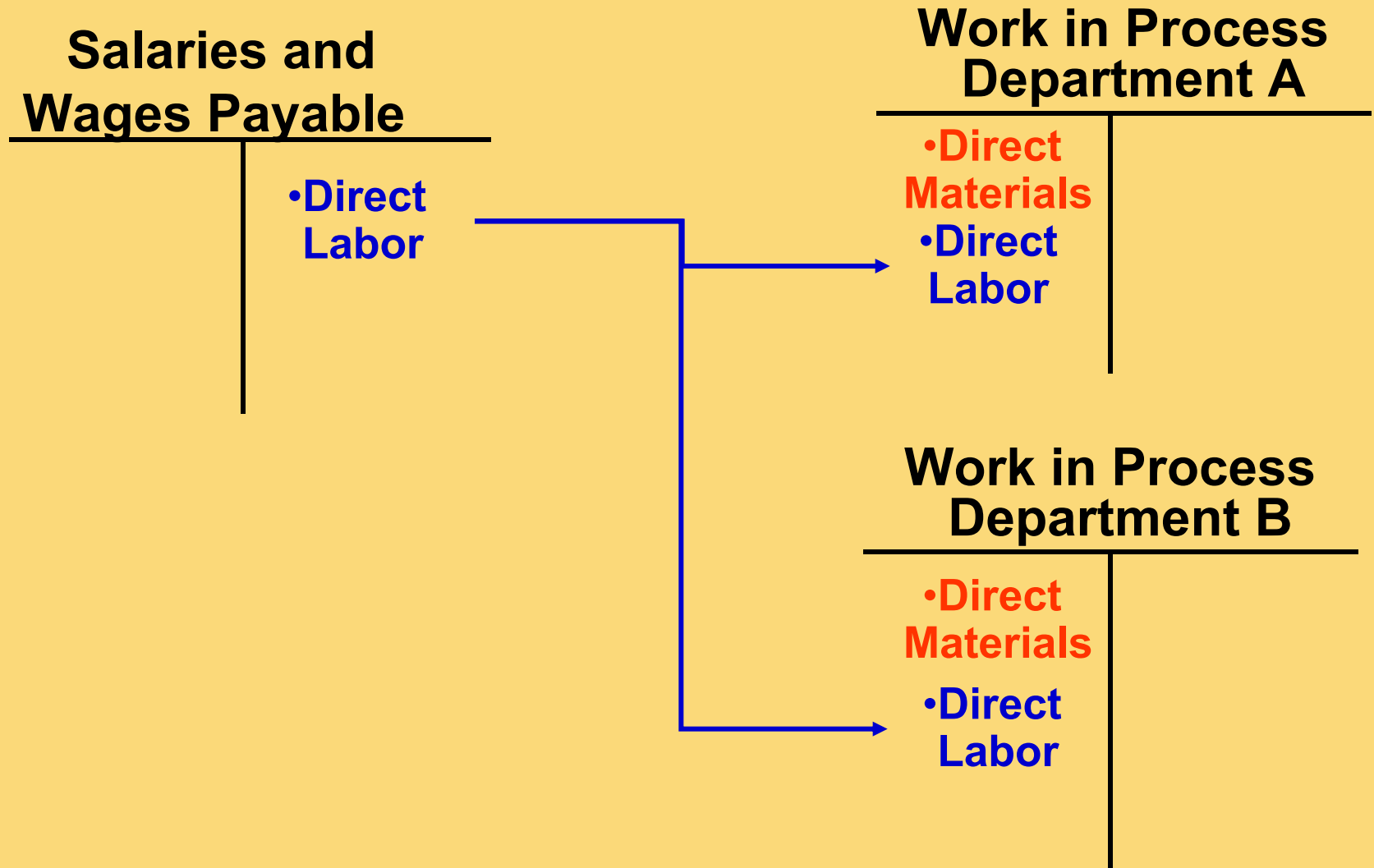
(in journal entry form)

GENERAL JOURNAL

Page 4

Date		Description	Post. Ref.	Debit	Credit
		Work in Process - Department A		XXXXXX	
		Work in Process - Department B		XXXXXX	
		Raw Materials			XXXXXX
		<i>To record the use of direct material.</i>			

Process Cost Flows (in T-account form)



Process Costing

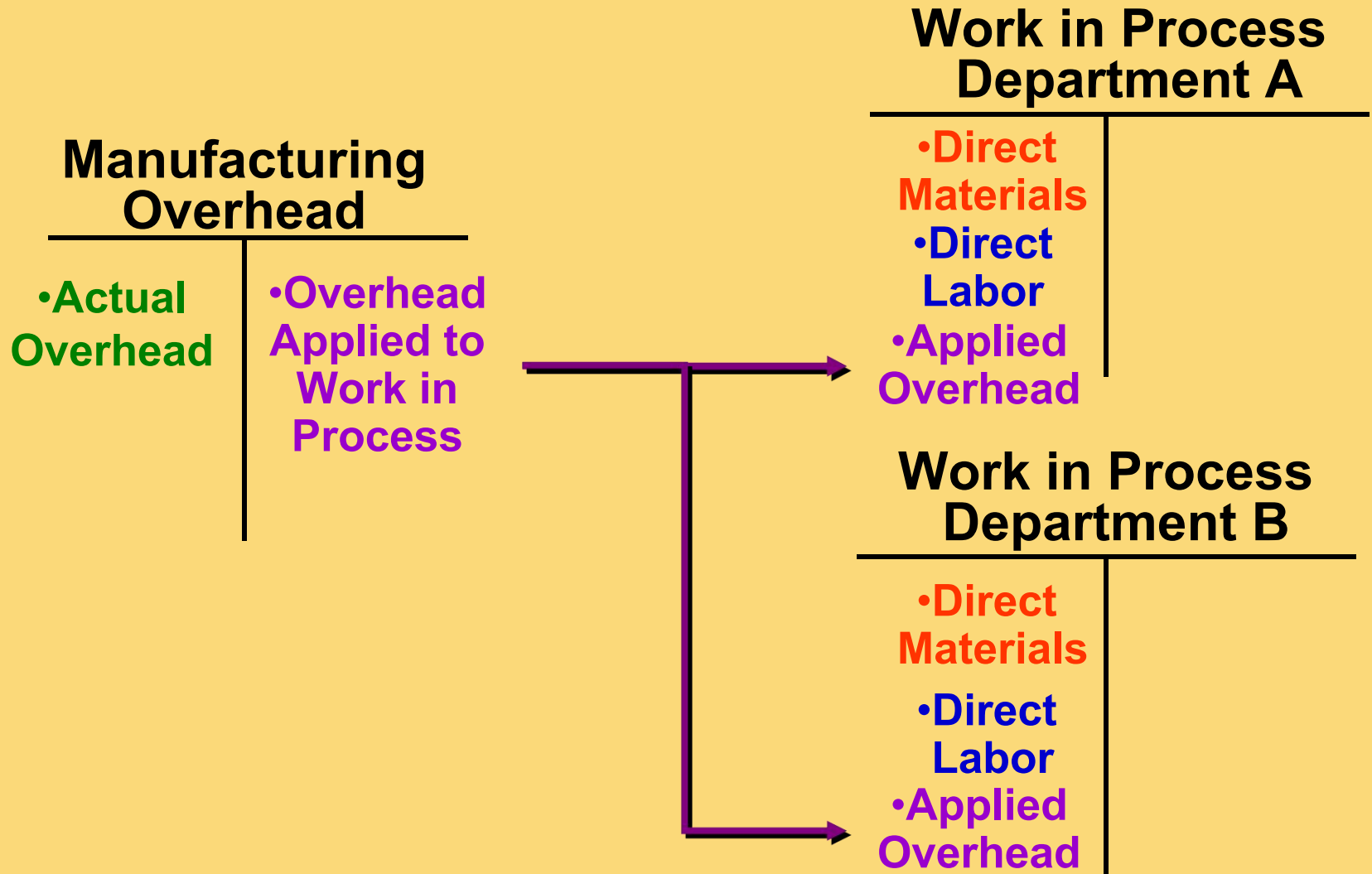
(in journal entry form)

GENERAL JOURNAL

Page 4

Date		Description	Post. Ref.	Debit	Credit
		Work in Process - Department A		XXXXXX	
		Work in Process - Department B		XXXXXX	
		Salaries and Wages Payable			XXXXXX
		<i>To record direct labor costs.</i>			

Process Cost Flows (in T-account form)



Process Costing

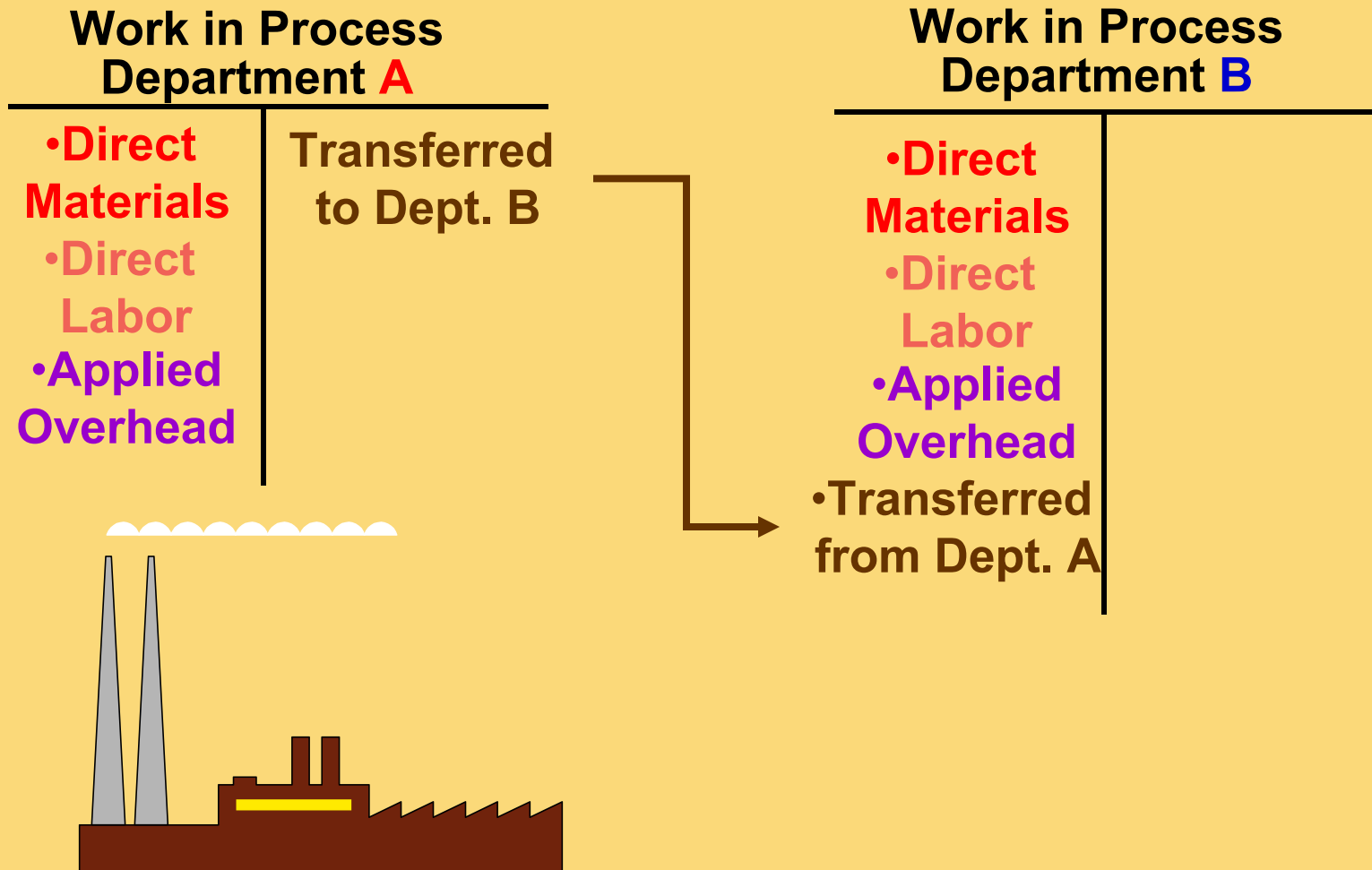
(In journal entry form)

GENERAL JOURNAL

Page 4

Date		Description	Post. Ref.	Debit	Credit
		Work in Process - Department A		XXXXXX	
		Work in Process - Department B		XXXXXX	
		Manufacturing Overhead			XXXXXX
		<i>To apply overhead to departments.</i>			

Process Cost Flows (in T-account form)



Process Costing

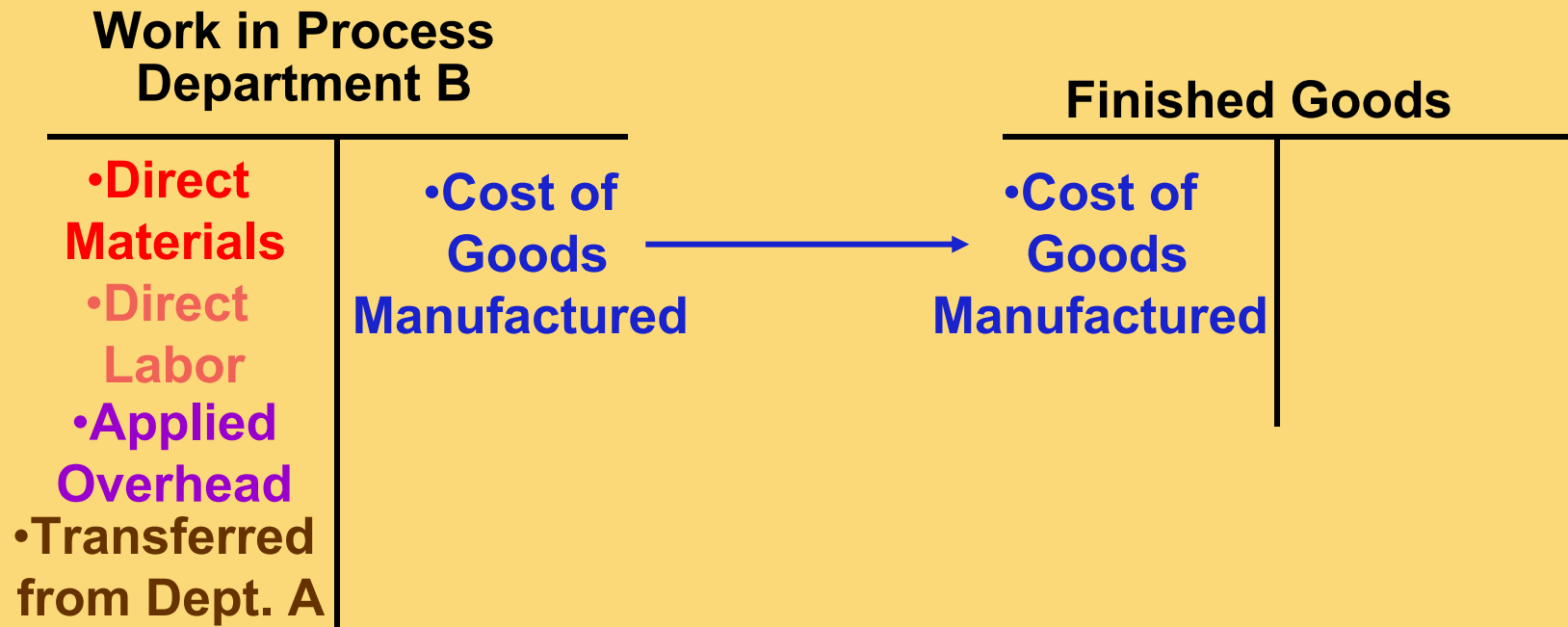
(in journal entry form)

GENERAL JOURNAL

Page 4

Date		Description	Post. Ref.	Debit	Credit
		Work in Process - Department B		XXXXXX	
		Work in Process - Department A			XXXXXX
		<i>To record the transfer of goods from</i>			
		<i>Department A to Department B.</i>			

Process Cost Flows (in T-account form)

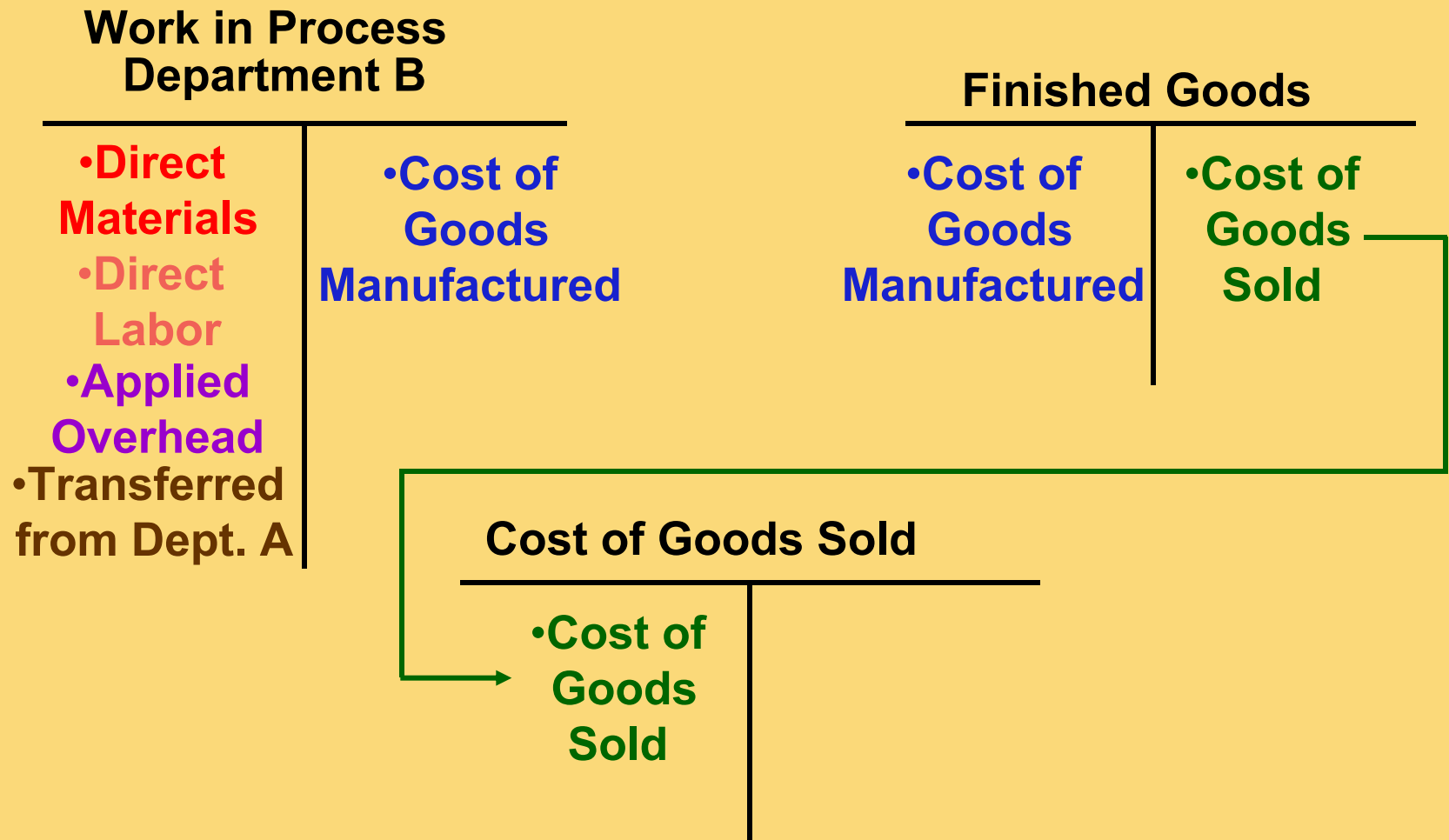


Process Costing

(in journal entry form)

GENERAL JOURNAL					Page 4
Date		Description	Post. Ref.	Debit	Credit
		Finished Goods		XXXXX	
		Work in Process - Department B			XXXXX
		<i>To record the completion of goods</i>			
		<i>and their transfer from Department B</i>			
		<i>to finished goods inventory.</i>			

Process Cost Flows (in journal entry form)



Process Costing

(in journal entry form)

GENERAL JOURNAL					Page 4
Date		Description	Post. Ref.	Debit	Credit
		Accounts Receivable		XXXXXX	
		Sales			XXXXXX
		<i>To record sales on account.</i>			
		Cost of Goods Sold		XXXXXX	
		Finished Goods			XXXXXX
		<i>To record cost of goods sold.</i>			

Product Cost Flows

<u>Raw Materials</u>	<u>Manufacturing Costs</u>	<u>Work In Process</u>
Beginning raw materials inventory	Direct materials	Beginning work in process inventory
+ Raw materials purchased	+ Direct labor	+ Total manufacturing costs
<u>= Raw materials available for use in production</u>	+ <u>Mfg. overhead</u>	<u>= Total work in process for the period</u>
- Ending raw materials	<u>= Total manufacturing costs</u>	- Ending work in process inventory
Costs associated with the goods that are completed during the period are transferred to finished goods inventory.		<u>= Cost of goods manufactured</u>



Equivalent Units of Production

Equivalent units are the product of the number of partially completed units and the percentage completion of those units.



We need to calculate equivalent units because a department usually has some partially completed units in its beginning and ending inventory.

Equivalent Units – The Basic Idea

Two half completed products are
equivalent to one completed product.



So, 10,000 units 70% complete
are ***equivalent to*** 7,000 complete units.

Quick Check ✓

For the current period, Jones started 15,000 units and completed 10,000 units, leaving 5,000 units in process 30 percent complete. How many equivalent units of production did Jones have for the period?

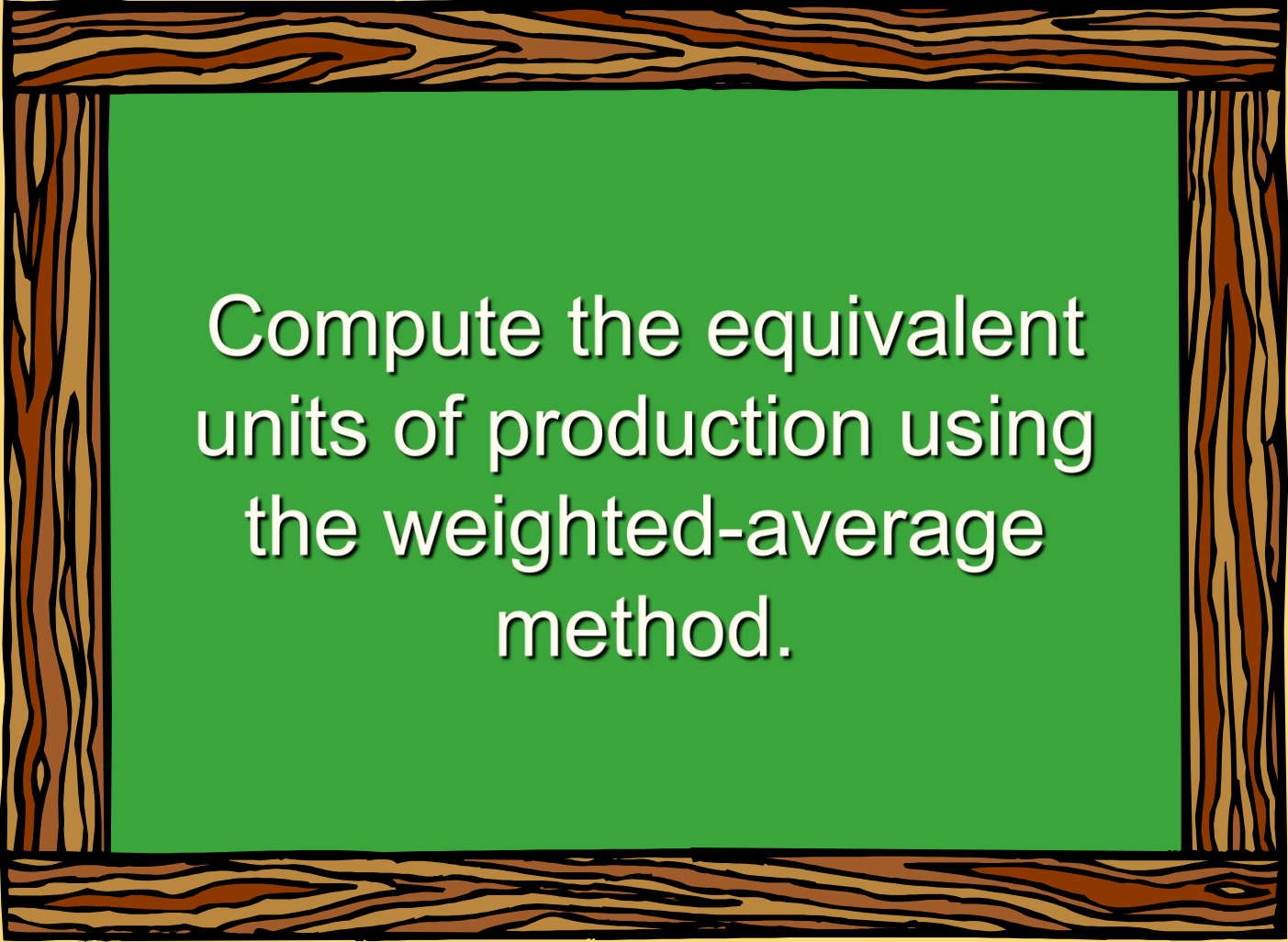
- a. 10,000
- b. 11,500
- c. 13,500
- d. 15,000

Calculating Equivalent Units

Equivalent units can be calculated two ways:

- ❶ The First-In, First-Out Method – FIFO is covered in the appendix to this chapter.
- ❷ The Weighted-Average Method – This method will be covered in the main portion of the chapter.

Learning Objective 2



Compute the equivalent units of production using the weighted-average method.

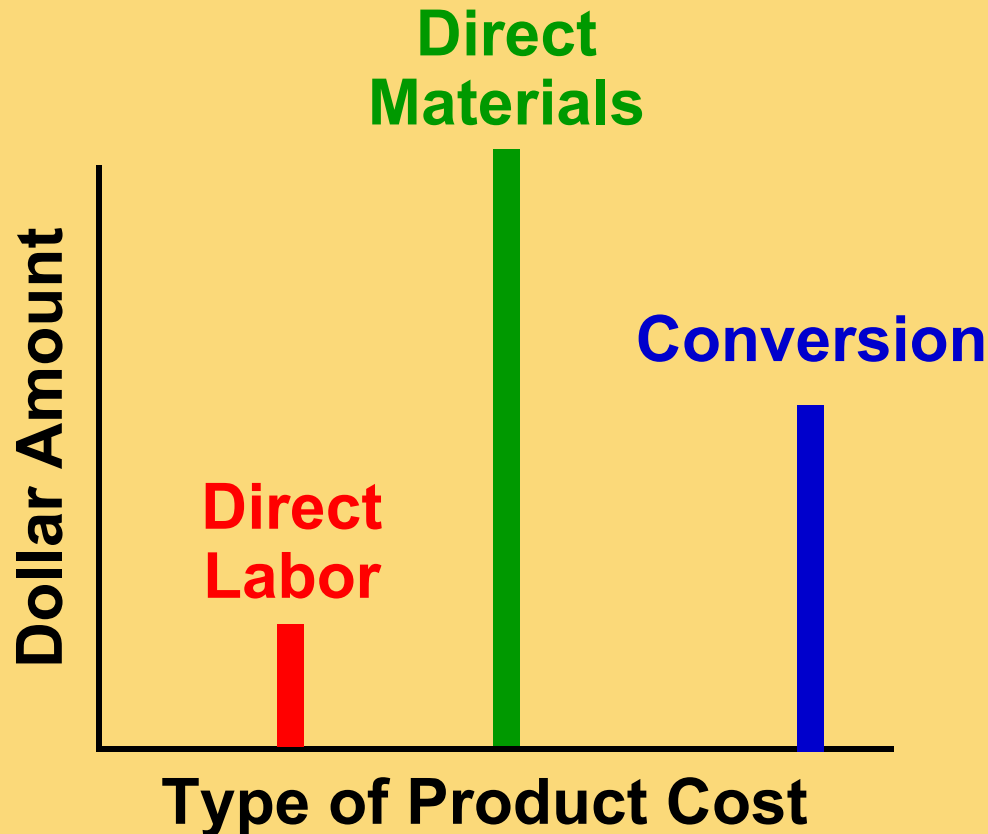
Equivalent Units of Production

Weighted-Average Method

The weighted-average method . . .

- Makes no distinction between work done in prior or current periods.
- Blends together units and costs from prior and current periods.
- Determines *equivalent units of production* for a department by adding together the number of units transferred out plus the *equivalent units in ending work in process inventory*.

Treatment of Direct Labor



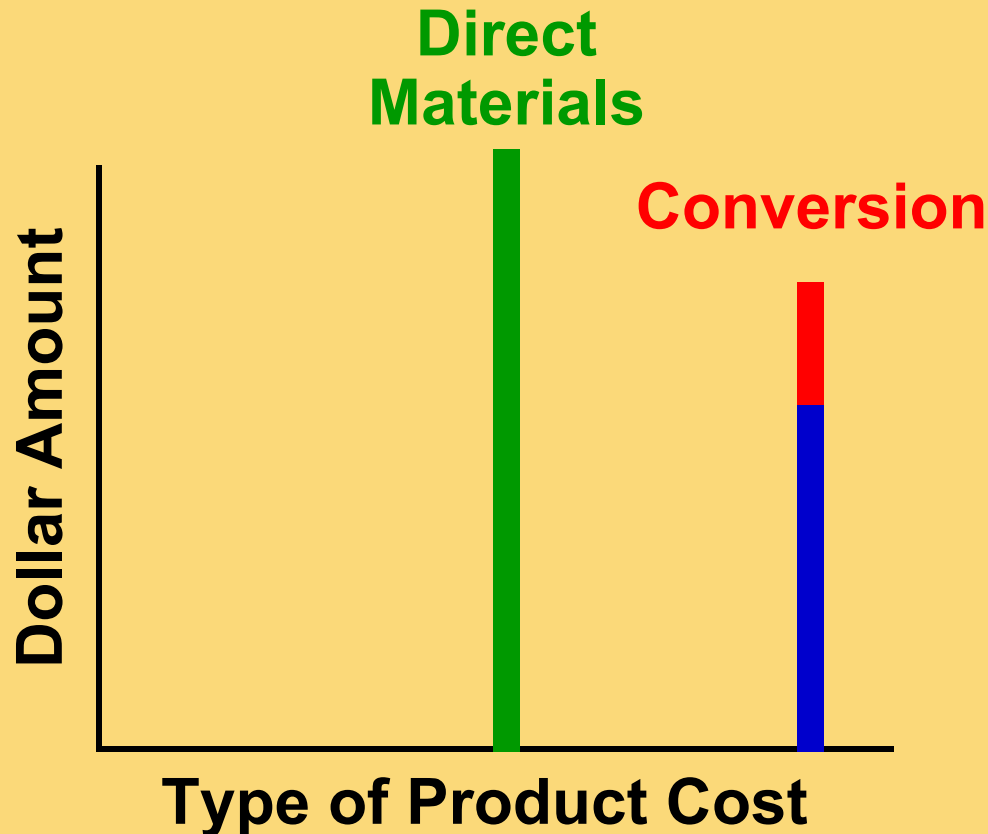
Direct labor costs may be small in comparison to other product costs in process cost systems.

Treatment of Direct Labor



One department of Nanjing Iron and Steel Group

Treatment of Direct Labor



Direct labor costs may be small in comparison to other product costs in process cost systems.

Direct labor and manufacturing overhead may be combined into one product cost called *conversion*.

Weighted-Average Example



Double Diamond Skis reported the following activity in Shaping and Milling Department for the month of May:

<u>Shaping and Milling Department</u>	<u>Units</u>	<u>Percent Completed</u>	
		<u>Materials</u>	<u>Conversion</u>
Beginning work in process	200	55%	30%
Units started into production in May	5,000		
Units completed during May and transferred to the next department	4,800	100%	100%
Ending work in process	400	40%	25%

Weighted-Average Example



The first step in calculating the equivalent units is to identify the units completed and transferred out of the Department in May (4,800 units)

	<u>Materials</u>	<u>Conversion</u>
Units completed and transferred to the next department	4,800	4,800

Weighted-Average Example



The second step is to identify the *equivalent units* of production in *ending work in process* with respect to *materials* for the month (160 units) and add this to the 4,800 units from step one.

	<u>Materials</u>	<u>Conversion</u>
Units completed and transferred to the next department	4,800	4,800
Work in process, June 30:		
400 units × 40%	160	
Equivalent units of Production in during the month of May	<u>4,960</u>	<u> </u>

Weighted-Average Example



The third step is to identify the *equivalent units* of production in *ending work in process* with respect to *conversion* for the month (100 units) and add this to the 4,800 units from step one.

	<u>Materials</u>	<u>Conversion</u>
Units completed and transferred to the next department	4,800	4,800
Work in process, June 30:		
400 units × 40%	160	
400 units × 25%		100
Equivalent units of Production in during the month of May	<u>4,960</u>	<u>4,900</u>

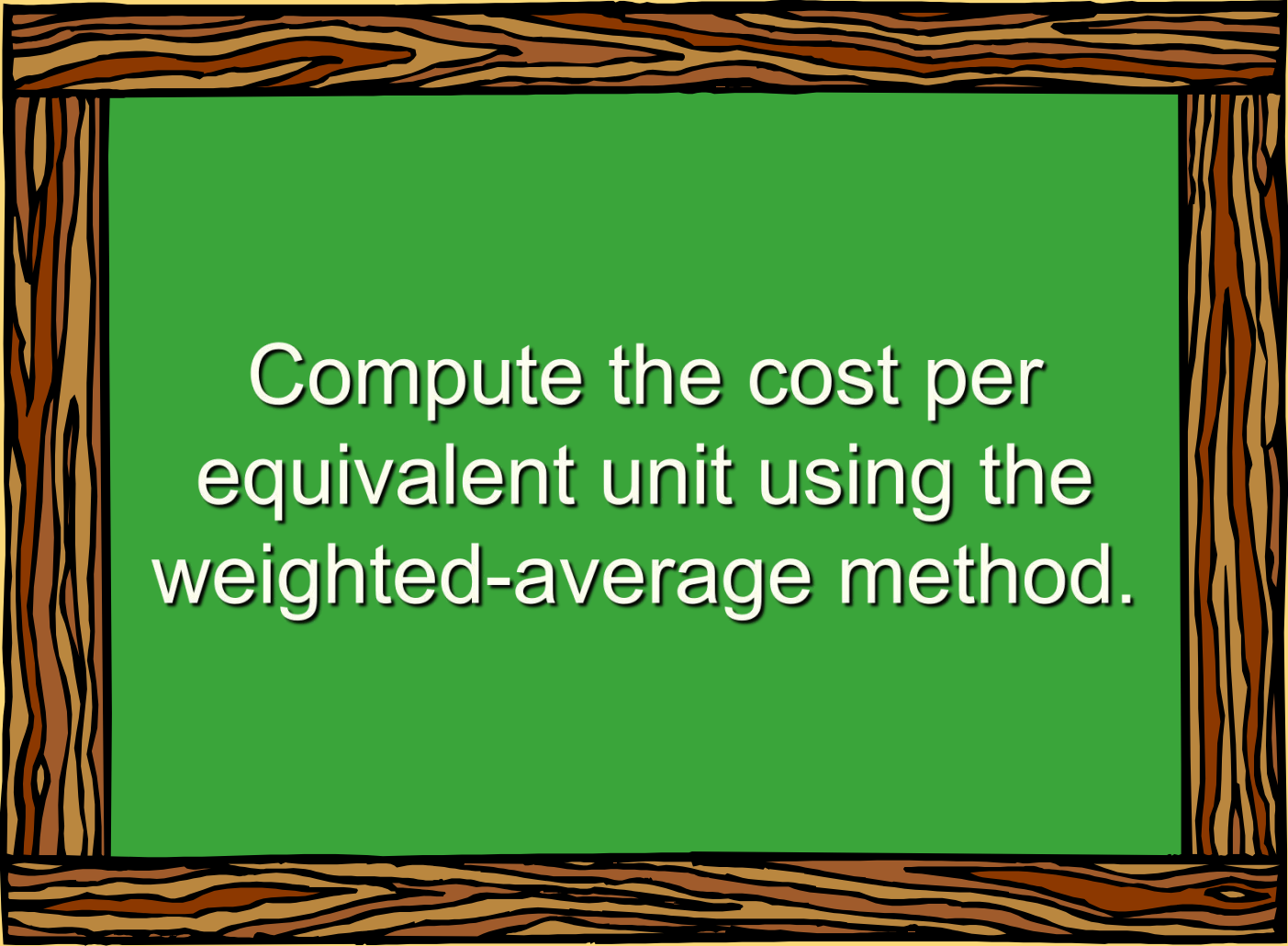
Weighted-Average Example



Equivalent units of production *always* equals:
Units completed and transferred
+ Equivalent units remaining in work in process

	<u>Materials</u>	<u>Conversion</u>
Units completed and transferred to the next department	4,800	4,800
Work in process, June 30:		
400 units × 40%	160	
400 units × 25%		100
Equivalent units of Production in during the month of May	4,960	4,900

Learning Objective 3



Compute the cost per equivalent unit using the weighted-average method.

Compute and Apply Costs



Beginning work in process:	200 units
Materials: 55% complete	\$ 9,600
Conversion: 30% complete	5,575
Production started during May	5,000 units
Production completed during May	4,800 units
Costs added to production in May	
Materials cost	\$ 368,600
Conversion cost	350,900
Ending work in process	400 units
Materials: 40% complete	
Conversion: 25% complete	

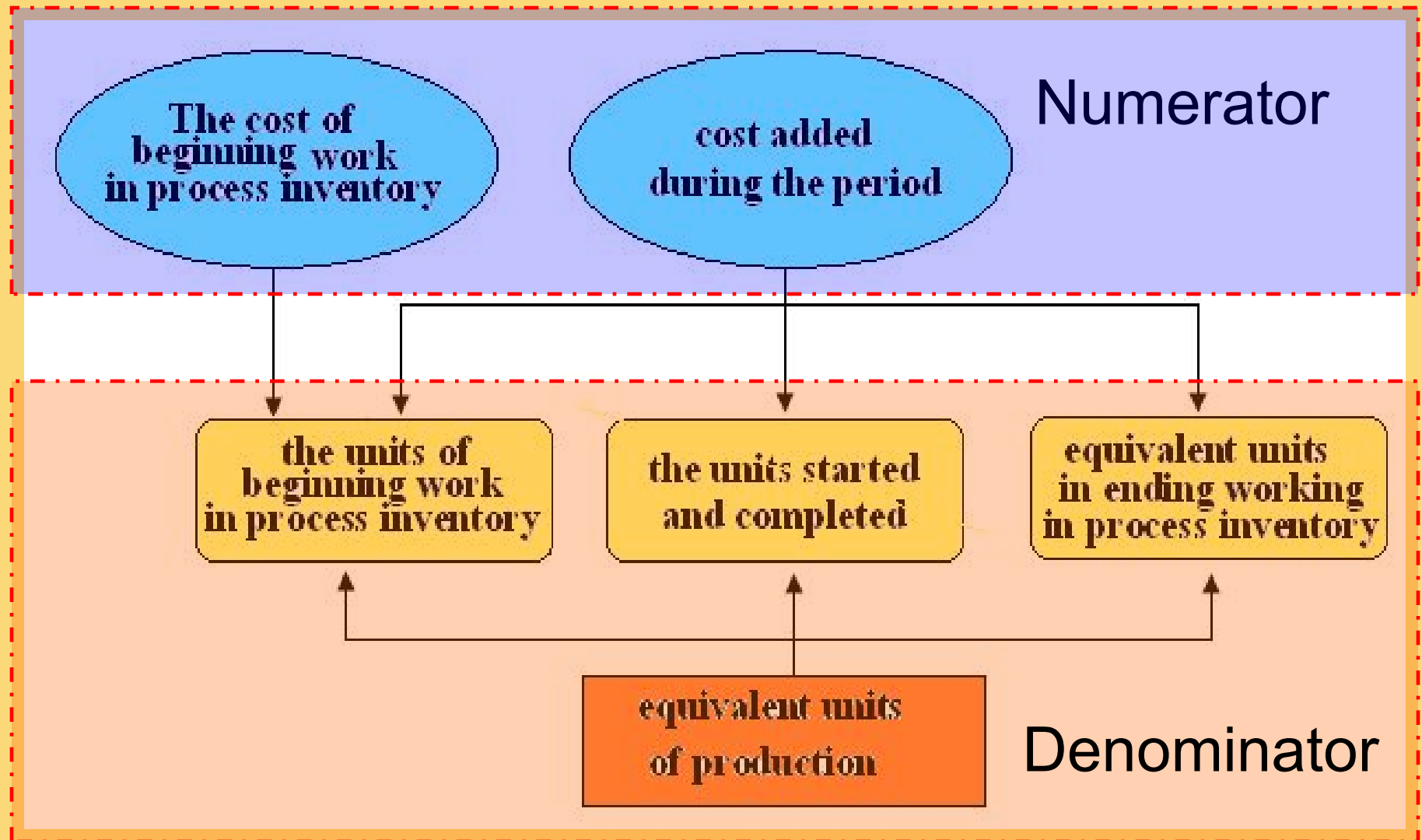
Compute and Apply Costs



The formula for computing the cost per equivalent unit is :

$$\text{Cost per equivalent unit} = \frac{\text{Cost of beginning work in process inventory} + \text{Cost added during the period}}{\text{Equivalent units of production}}$$





Compute and Apply Costs



Here is a schedule with the cost and equivalent unit information.

	<u>Total Cost</u>	<u>Materials</u>	<u>Conversion</u>
Cost to be accounted for:			
Work in process, May 1	\$ 15,175	\$ 9,600	\$ 5,575
Costs added in the Shipping and Milling Department	719,500	368,600	350,900
Total cost	<u>\$ 734,675</u>	<u>\$ 378,200</u>	<u>\$ 356,475</u>
 Equivalent units		4,960	4,900
Cost per equivalent unit			

Compute and Apply Costs

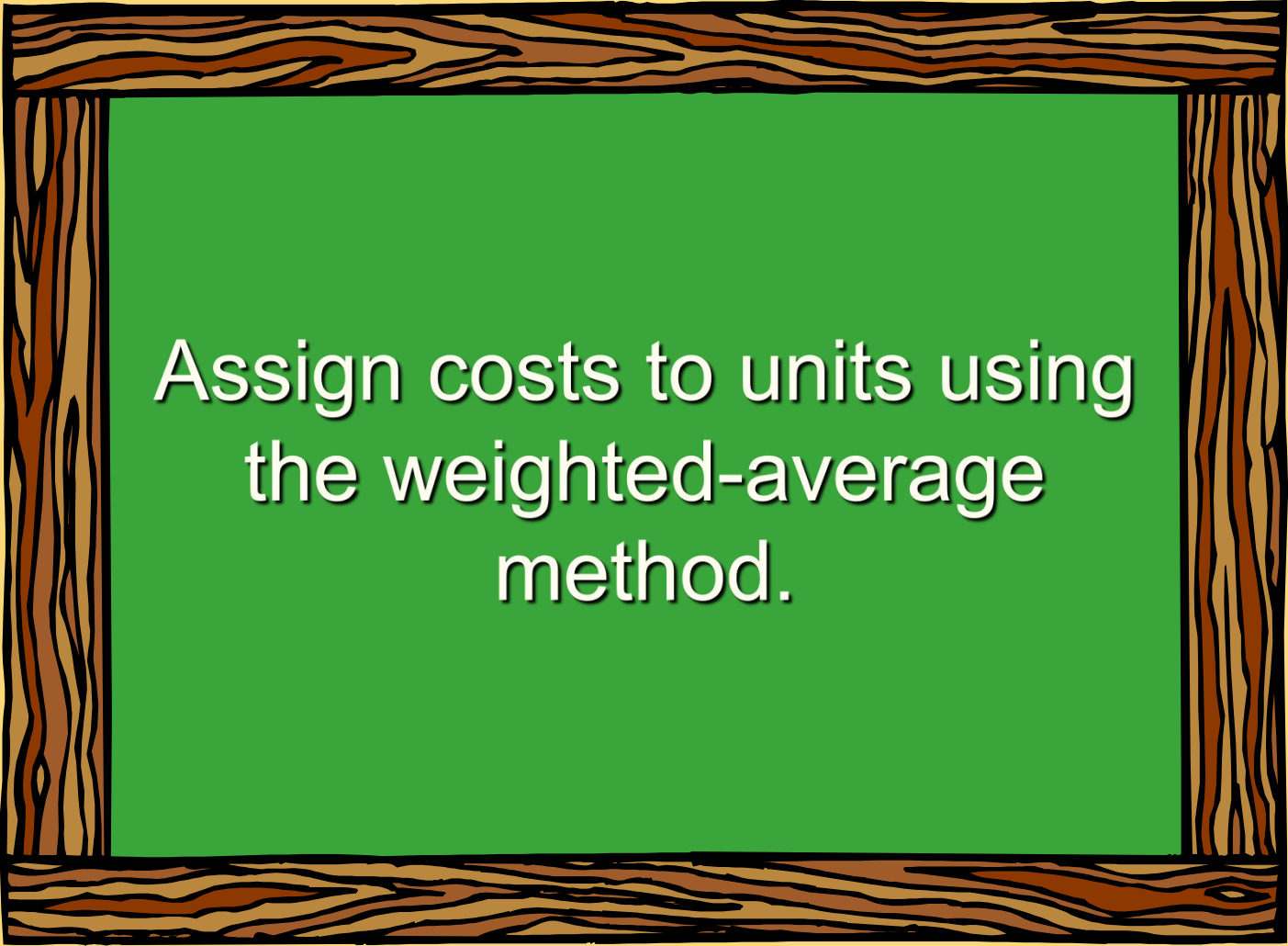


Here is a schedule with the cost and equivalent unit information.

	<u>Total Cost</u>	<u>Materials</u>	<u>Conversion</u>
Cost to be accounted for:			
Work in process, May 1	\$ 15,175	\$ 9,600	\$ 5,575
Costs added in the Shipping and Milling Department	719,500	368,600	350,900
Total cost	<u><u>\$ 734,675</u></u>	<u><u>\$ 378,200</u></u>	<u><u>\$ 356,475</u></u>
Equivalent units		4,960	4,900
Cost per equivalent unit		\$ 76.25	\$ 72.75
Total cost per equivalent unit = \$76.25 + \$72.75 = \$149.00			

\$35 $\$378,200 \div 4,960 \text{ units} = \76.25

Learning Objective 4



Assign costs to units using
the weighted-average
method.

Basic Equation for Inventory Accounts

**Beginning
balance**

+

**Additions
to inventory**

=

**Ending
balance**

+

**Withdrawals
from
inventory**



Applying Costs



Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	



Applying Costs



Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	
Cost per equivalent unit	\$ 76.25	\$ 72.75	



Applying Costs



Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	
Cost per equivalent unit	\$ 76.25	\$ 72.75	
Cost of ending work in process inventory	<u>\$ 12,200</u>	<u>\$ 7,275</u>	\$ 19,475



Computing the Cost of Units Transferred Out

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	
Cost per equivalent unit	\$ 76.25	\$ 72.75	
Cost of ending work in process inventory	<u>\$ 12,200</u>	<u>\$ 7,275</u>	\$ 19,475
Units completed and transferred out:			
Units transferred to the next department	4,800	4,800	



Computing the Cost of Units Transferred Out

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	Materials	Conversion	Total
Ending work in process inventory:			
Equivalent units of production	160	100	
Cost per equivalent unit	\$ 76.25	\$ 72.75	
Cost of ending work in process inventory	\$ 12,200	\$ 7,275	\$ 19,475
Units completed and transferred out:			
Units transferred to the next department	4,800	4,800	
Cost per equivalent unit	\$ 76.25	\$ 72.75	



Computing the Cost of Units Transferred Out

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	
Cost per equivalent unit	\$ 76.25	\$ 72.75	
Cost of ending work in process inventory	<u>\$ 12,200</u>	<u>\$ 7,275</u>	\$ 19,475
Units completed and transferred out:			
Units transferred to the next department	4,800	4,800	
Cost per equivalent unit	\$ 76.25	\$ 72.75	
Cost of units transferred out	<u>\$ 366,000</u>	<u>\$ 349,200</u>	\$ 715,200



Reconciling Costs



Shaping and Milling Department Cost Reconciliation

Costs to be accounted for:

Cost of beginning work in process inventory (shown on slide 39)	\$	15,175
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Costs added to production during the period (shown on slide 39)		719,500
--	--	---------

Total cost to be accounted for	<u>\$</u>	<u>734,675</u>
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Reconciling Costs



Shaping and Milling Department Cost Reconciliation

Costs to be accounted for:

Cost of beginning work in process inventory	\$	15,175
Costs added to production during the period		719,500
Total cost to be accounted for	\$	<u>734,675</u>

Cost accounted for as follows:

Cost of ending work in process inventory	\$	19,475
Cost of units transferred out		715,200
Total cost accounted for	\$	<u>734,675</u>

FIFO Method



Appendix 4A

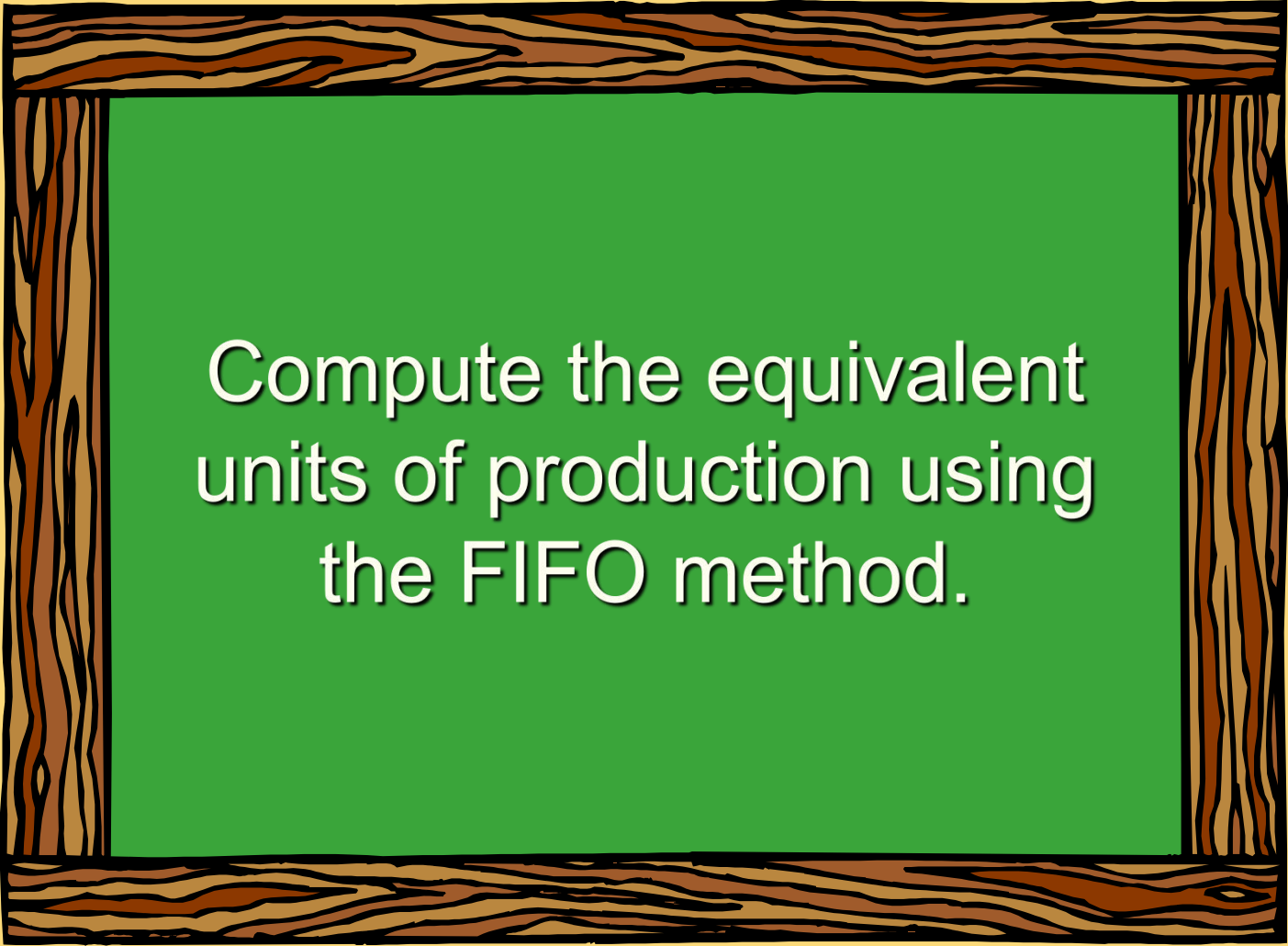


FIFO vs. Weighted-Average Method

The FIFO method (generally considered more accurate than the weighted-average method) differs from the weighted-average method in two ways:

1. The computation of equivalent units.
2. The way in which the costs of beginning inventory are treated in the cost reconciliation report.

Learning Objective 5



Compute the equivalent units of production using the FIFO method.

Equivalent Units – FIFO Method



Let's revisit the Double Diamond Skis example.
Assume the following activity is reported in
Shaping and Milling Department for May:

<u>Shaping and Milling Department</u>	<u>Units</u>	<u>Percent Completed</u>	
		<u>Materials</u>	<u>Conversion</u>
Beginning work in process	200	55%	30%
Units started into production in May	5,000		
Units completed during May and transferred to the next department	4,800	100%	100%
Ending work in process	400	40%	25%

Equivalent Units – FIFO Method



Step 1: Determine equivalent units needed to complete beginning inventory.

	<u>Materials</u>	<u>Conversion</u>
To complete beginning work in process:		
Materials: 200 units $\times$ (100% - 55%)	90	
Conversion: 200 units $\times$ (100% - 30%)		140

Equivalent Units – FIFO Method



Step 2: Determine units started and completed during the period.

	<u>Materials</u>	<u>Conversion</u>
To complete beginning work in process:		
Materials: 200 units $\times$ (100% - 55%)	90	
Conversion: 200 units $\times$ (100% - 30%)		140
Units started and completed during May	4,600	4,600

Equivalent Units – FIFO Method



Step 3: Add the equivalent units in ending work in process inventory.

	<u>Materials</u>	<u>Conversion</u>
To complete beginning work in process:		
Materials: 200 units × (100% - 55%)	90	
Conversion: 200 units × (100% - 30%)		140
Units started and completed during May	4,600	4,600
Ending work in process		
Materials: 400 units × 40% complete	160	
Conversion: 400 units × 25% complete		100
Equivalent units of production	<u>4,850</u>	<u>4,840</u>

FIFO Example



Materials

5,000 Units Started

**Beginning
Work in Process
200 Units
55% Complete**

**4,600 Units Started
and Completed**

**Ending
Work in Process
400 Units
40% Complete**

**200 ×
45%**

**90 Equivalent Units
4,600 Units Completed
160 Equivalent Units
4,850 Equivalent units
of production**

**400 ×
40%**

FIFO Example



Conversion

5,000 Units Started

Beginning
Work in Process
200 Units
30% Complete

4,600 Units Started
and Completed

Ending
Work in Process
400 Units
25% Complete

200 ×
70%

140 Equivalent Units
4,600 Units Completed
100 Equivalent Units
4,840 Equivalent units
of production

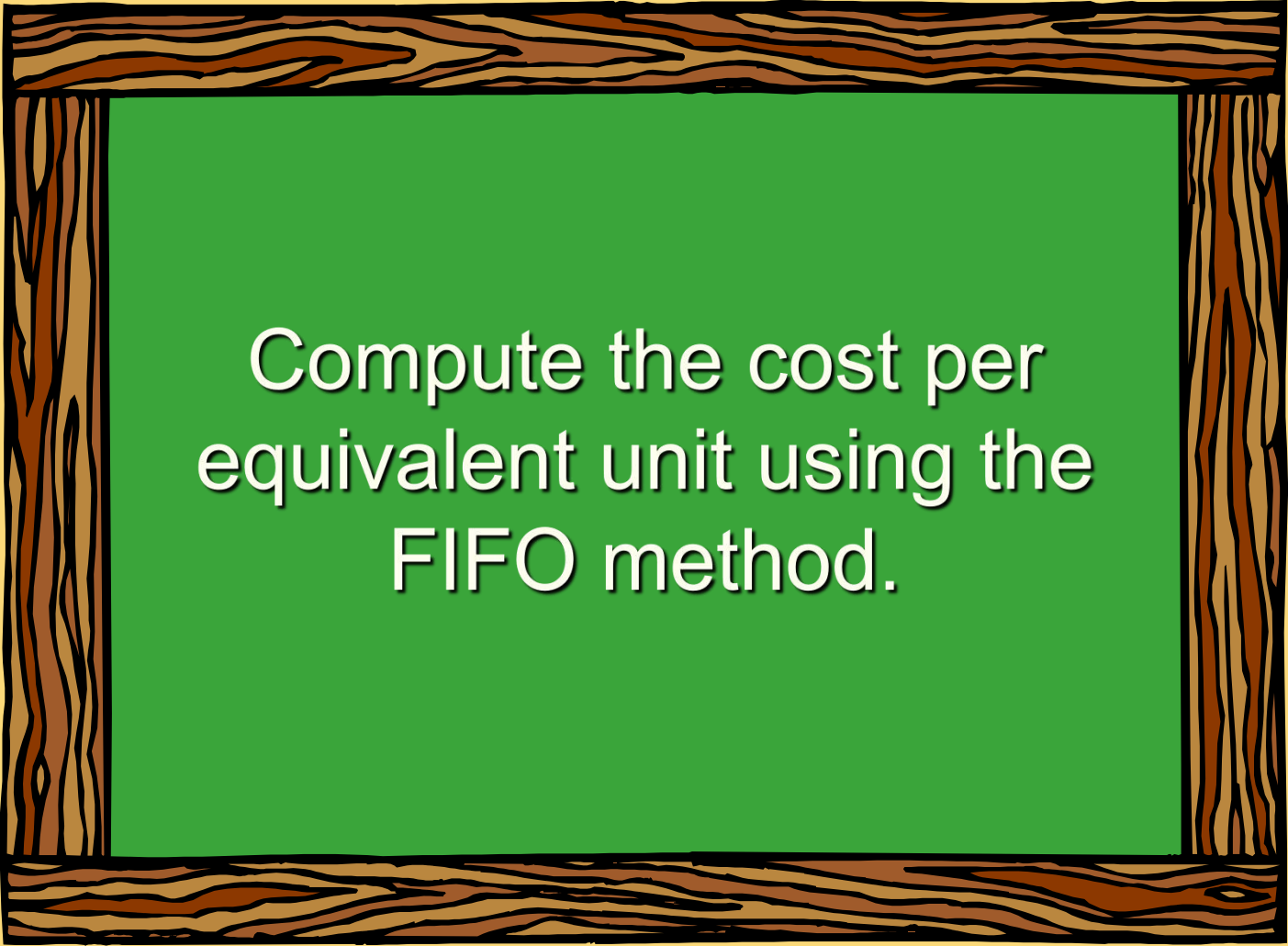
400 ×
25%

Equivalent Units: Weighted Average vs. FIFO

As shown below, the equivalent units in beginning inventory are subtracted from the equivalent units of production per the weighted-average method to obtain the equivalent units of production under the FIFO method.

	<u>Materials</u>	<u>Conversion</u>
Equivalent units - weighted average method	4,960	4,900
Less equivalent units in beginning inventory:		
200 units × 55%	110	
200 units × 30%		60
Equivalent units - FIFO method	<u>4,850</u>	<u>4,840</u>

Learning Objective 6



Compute the cost per equivalent unit using the FIFO method.

Cost per Equivalent Unit - FIFO



Let's revisit the Double Diamond Skis Shaping and Milling Department for the Month of May to prepare our production report.

Beginning work in process: **200 units**

Materials: 55% complete **\$ 9,600**

Conversion: 30% complete **5,575**

\$15, 175

Production started during May **5,000 units**

Production completed during May **4,800 units**

Costs added to production in May

Materials cost **\$ 368,600**

Conversion cost **350,900**

Ending work in process: **400 units**

Materials: 40% complete

Conversion: 25% complete

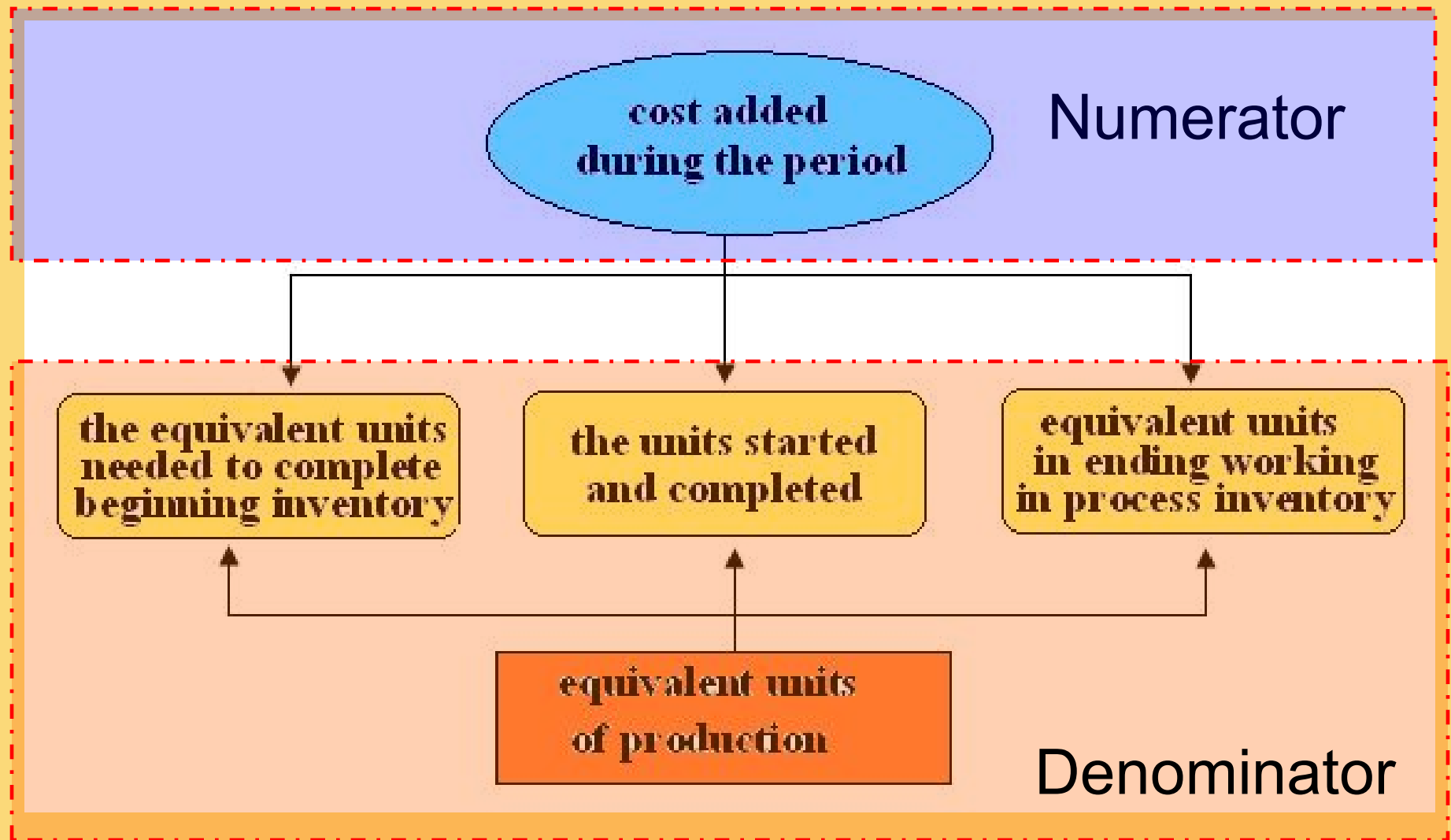
Cost per Equivalent Unit - FIFO



The formula for computing the cost per equivalent unit under FIFO method is as follows:

$$\text{Cost per equivalent unit} = \frac{\text{Cost added during the period}}{\text{Equivalent units of production}}$$





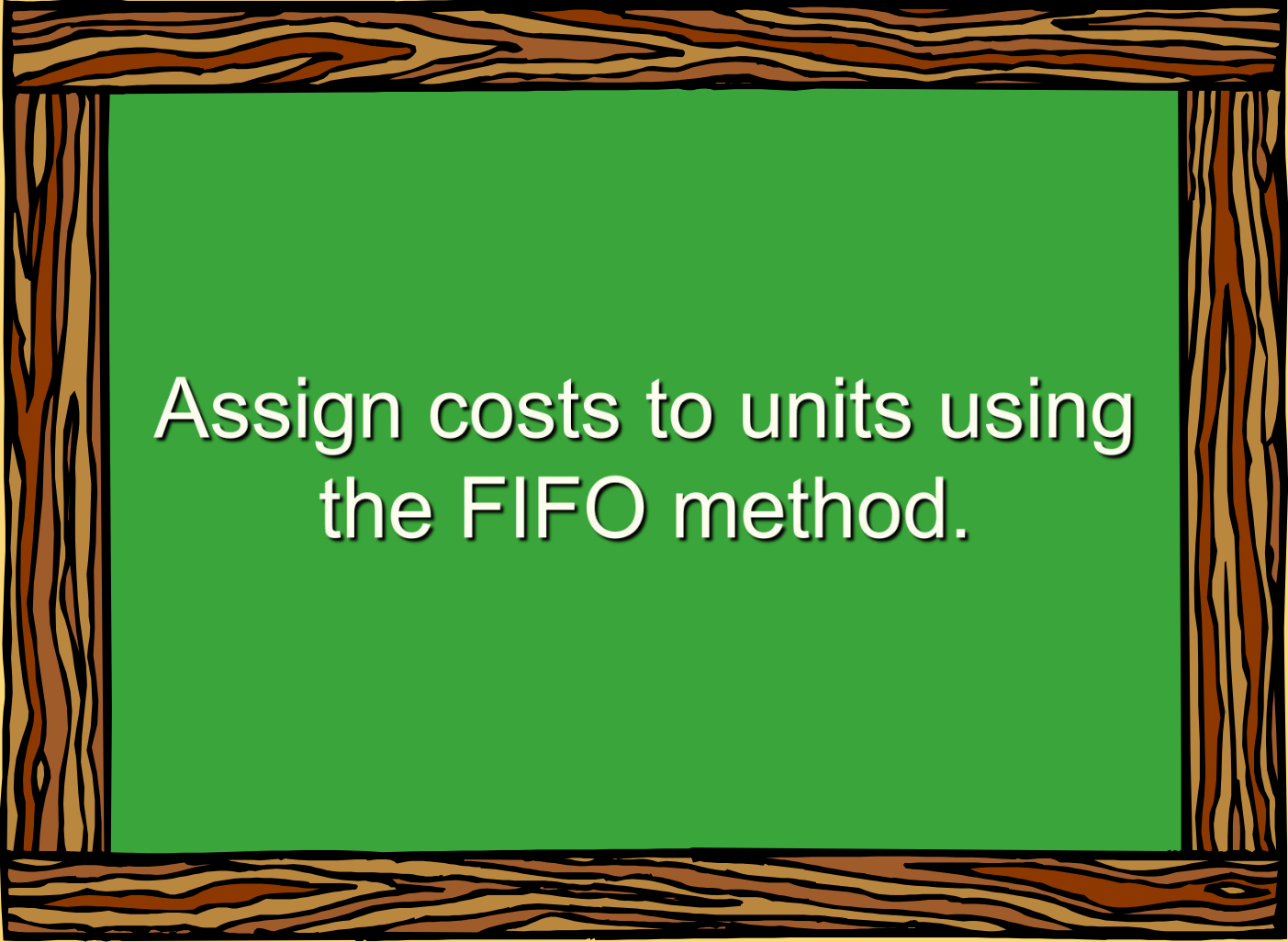
Cost per Equivalent Unit - FIFO



	<u>Total Cost</u>	<u>Materials</u>	<u>Conversion</u>
Costs added in the Shaping and Milling Department	\$ 719,500	\$ 368,600	\$ 350,900
Equivalent units		4,850	4,840
Cost per equivalent unit		\$ 76.00	\$ 72.50
Total cost per equivalent unit = \$76.00 + \$72.50 = \$148.50			



Learning Objective 7



Assign costs to units using
the FIFO method.

Applying Costs - FIFO



Step 1: Record the equivalent units of production in ending work in process inventory.

Shaping and Milling Department Cost of Ending Work in Porcess Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	

Applying Costs - FIFO



Step 2: Record the cost per equivalent unit.

Shaping and Milling Department Cost of Ending Work in Porcess Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	
Cost per equivalent unit	\$ 76.00	\$ 72.50	

Applying Costs - FIFO



Step 3: Compute the cost of ending work in process inventory.

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Ending work in process inventory:			
Equivalent units of production	160	100	
Cost per equivalent unit	\$ 76.00	\$ 72.50	
Cost of ending work in process inventory	<u>\$ 12,160</u>	<u>\$ 7,250</u>	\$ 19,410

Cost of Units Transferred Out



Step 1: Record the cost in beginning work in process inventory.

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Units transferred out:			
Cost in beginning work in process (WIP)	\$ 9,600	\$ 5,575	\$ 15,175

Cost of Units Transferred Out



Step 2: Compute the cost to complete the units in beginning work in process inventory.

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	<u>Materials</u>	<u>Conversion</u>	<u>Total</u>
Units transferred out:			
Cost in beginning work in process (WIP)	\$ 9,600	\$ 5,575	\$ 15,175
Cost to complete beginning WIP:			
Equivalent units to complete	90	140	
Cost per equivalent unit	\$ 76.00	\$ 72.50	
Cost to complete beginning WIP	\$ 6,840	\$ 10,150	\$ 16,990

Cost of Units Transferred Out



Step 3: Compute the cost of units started and completed this period.

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	Materials	Conversion	Total
Units transferred out:			
Cost in beginning work in process (WIP)	\$ 9,600	\$ 5,575	\$ 15,175
Cost to complete beginning WIP:			
Equivalent units to complete	90	140	
Cost per equivalent unit	\$ 76.00	\$ 72.50	
Cost to complete beginning WIP	\$ 6,840	\$ 10,150	\$ 16,990
Cost of units started and completed in May			
Units started and completed in May	4,600	4,600	
Cost per equivalent unit	\$ 76.00	\$ 72.50	
Cost of units started and completed in May	\$ 349,600	\$ 333,500	\$ 683,100
Cost of units transferred out			

Cost of Units Transferred Out



Step 4: Compute the total cost of units transferred out.

Shaping and Milling Department Cost of Ending Work in Process Inventory and the Units Transferred Out

	Materials	Conversion	Total
Units transferred out:			
Cost in beginning work in process (WIP)	\$ 9,600	\$ 5,575	\$ 15,175
Cost to complete beginning WIP:			
Equivalent units to complete	90	140	
Cost per equivalent unit	\$ 76.00	\$ 72.50	
Cost to complete beginning WIP	\$ 6,840	\$ 10,150	\$ 16,990
Cost of units started and completed in May			
Units started and completed in May	4,600	4,600	
Cost per equivalent unit	\$ 76.00	\$ 72.50	
Cost of units started and completed in May	\$ 349,600	\$ 333,500	\$ 683,100
Cost of units transferred out			<u><u>\$ 715,265</u></u>

Reconciling Costs



Shaping and Milling Department Cost Reconciliation

Costs to be accounted for:

Cost of beginning work in process inventory	\$ 15,175
Costs added to production during the period	719,500
Total cost to be accounted for	<u>\$ 734,675</u>

Reconciling Costs



Shaping and Milling Department Cost Reconciliation

Costs to be accounted for:

Cost of beginning work in process inventory	\$ 15,175
Costs added to production during the period	719,500
Total cost to be accounted for	<u>\$ 734,675</u>

Cost accounted for as follows:

Cost of ending work in process inventory	\$ 19,410
Cost of units transferred out	715,265
Total cost accounted for	<u>\$ 734,675</u>

End of Chapter 4

